

Introduction

Generative AI provides malicious actors with means to socially manipulate and extort users in new, hard to prevent ways [1] [2]. Existing methods of URL phishing prevention have been trained and built to detect URLs generated without the use of AI, with only ~1-4% of URLs beating the detection methods [3]. New nefarious AI powered models like “DeepPhish” are able to generate URLs which beat existing detection systems ~20–35% of the time [3]. These results are accepted, and being actively mitigated by security teams around the world. Our research team’s goal however is to understand the “threat factors” which generative AI models are using to generate this observed outperformance against existing detection methods. Threat factors and existing methods of creating nefarious URLs are well researched [4][5], and we want to determine which threat factors malicious generative AI systems are relying on in hopes of enabling better detection methods.

Motivation

Phishing complaints and related economic losses are rapidly increasing in the United States of America. The FBI reported a 208% increase in year over year losses (\$70 million in 2024 and \$215.8 million in 2025), an 1100% increase over the last two years (\$18.7 million in 2023), and a \$3.05 billion annual loss in 2025 due to business email compromises (BEC), which often start with a phishing attack [6]. According to KnowBe4, as of March 2025 AI was being utilized in 82.6% of phishing attacks, representing a 53.5% year over year increase [7]. Phishing has always been around, but generative AI is enabling malicious actors to utilize social engineering in scalable and highly effective ways to manipulate and harm targeted, and often already vulnerable, individuals [8] [9]. Malicious URLs are one of the computationally cheapest and easiest ways to detect malicious phishing emails due to the small size of text being evaluated [4]. By identifying the most prominent URL threat factors AI models are using to generate outperformance against prior detection models, we hope to generate information which could enable security researchers to improve existing detection methods and build new detection methods in a cost-effective manner.

Sources

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